FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4

MATHEMATICS

CONVEX OPTIMIZATION

Module IV: Duality

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	For the problem minimize $f_0(x)$ subject to $f_i(x) \leq 0$ and $h_j(x) = 0$, define the Lagrangian and the Lagrange dual function. State the sign restrictions on the dual variables.	**3**
2	Show that the Lagrange dual function is concave, even when the primal problem is not convex.	**3**
3	Define the optimal duality gap. Distinguish between weak duality and strong duality. How can a primal feasible point and a dual feasible point certify optimality?	**3**
4	Write the inequalities defining a saddle point $(x^\star, \lambda^\star, \nu^\star)$ of a Lagrangian, with $\lambda \geq 0$. Which variables are minimized and which are maximized?	**3**
5	What is complementary slackness of a dual canonical linear programming problem ?	**3**
6	State Slater's condition for a convex problem with inequalities $f_i(x) \leq 0$ and equalities $Ax = b$. What does it imply about strong duality and dual attainment when the primal optimum is finite?	**3**
7	List the four groups of KKT conditions for a differentiable problem. State when these conditions are sufficient for global optimality and a constraint qualification under which they are also necessary.	**3**
8	Distinguish between weak alternatives and strong alternatives. State an alternative system that certifies infeasibility of $Ax \leq b$, where inequalities are componentwise.	**3**
9	For a constraint $f(x) \preceq_K 0$, in which cone must its Lagrange multiplier lie? Explain why its contribution to the Lagrangian is nonpositive at a feasible point.	**3**
10	For a semidefinite constraint $F(x) \preceq 0$ with multiplier $Z \succeq 0$, write the complementary slackness condition. Give an example with both $F(x)$ and Z nonzero that satisfies this condition.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Find the dual function of the LP minimize $c^T x$ subject to $Gx \leq h$, $Ax = b$, where $x \in \mathbb{R}^n$. Give the dual problem, and make the implicit equality constraints explicit.	**6**
12	Consider minimize x subject to $x^2 \leq 0$, with $x \in \mathbb{R}$. Sketch $\mathcal{G} = \{(x^2, x) : x \in \mathbb{R}\}$ and $\mathcal{A} = \{(u,t) : \exists x,\ x^2 \leq u,\ x \leq t\}$. Derive the dual and find the primal and dual optimal values. Discuss dual attainment, Slater's condition and strong duality.	**6**
13	Prove weak duality for a standard-form optimization problem without assuming convexity. If x is primal feasible and (λ, ν) is dual feasible with finite $g(\lambda, \nu)$, show that $0 \leq f_0(x) - p^\star \leq f_0(x) - g(\lambda, \nu)$, assuming $p^\star$ is finite.	**6**
14	Prove that a pair of feasible solutions of dual canonical linear programming problems exhibit complementary slackness if and only if they are optimal solutions.	**6**
15	Consider minimize $\|Ax - b\|_2^2$ subject to $Gx = h$, where $A \in \mathbb{R}^{m \times n}$ has rank n and $G \in \mathbb{R}^{p \times n}$ has rank p. Give the KKT conditions, and derive expressions for the primal solution $x^\star$ and equality multiplier $\nu^\star$.	**6**
16	State and prove Farkas' lemma in the following form: the systems $Ax \leq 0$, $c^T x < 0$ and $A^T y + c = 0$, $y \geq 0$ are strong alternatives. You may use strong duality for linear programming.	**6**
17	Find the Lagrange dual of minimize $c^T x$ subject to $Ax \preceq_K b$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $K \subseteq \mathbb{R}^m$ is a proper cone. Make any implicit equality constraints explicit.	**6**
18	Derive the dual of the SDP minimize $c^T x$ subject to $F(x) = G + \sum_{i=1}^n x_i F_i \preceq 0$, where $F_i, G \in \mathbb{S}^k$. Assuming the primal optimal value is finite, give a sufficient strict-feasibility condition for strong duality.	**6**

Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Consider minimize $x^2 + 1$ subject to $(x-2)(x-4) \leq 0$, with $x \in \mathbb{R}$. (a) Find the feasible set and primal optimum. Derive the dual function and solve the dual problem. **(5 marks)** (b) Verify the KKT conditions at the optimal primal-dual pair. Show directly that this pair is a saddle point of the Lagrangian, and identify the common primal and dual optimal value. **(5 marks)**	**10**
20	(a) For a standard-form problem with domain $\mathbb{R}^n$, prove that a finite saddle point of the Lagrangian over $x \in \mathbb{R}^n$, $\lambda \geq 0$, $\nu \in \mathbb{R}^p$ gives primal and dual optimal solutions with zero duality gap. Do not assume convexity. **(5 marks)** (b) For minimize $f_0(x)$ subject to $f_i(x) \preceq_{K_i} 0$, $Ax = b$, formulate the dual using K_i^*. Prove weak duality and derive the generalized complementary slackness conditions when finite equal primal and dual optima are attained. **(5 marks)**	**10**